

Ko'p o'zgaruvchili funktsiyaning ekstremumlari

Reja

1⁰. Funktsiya ekstremumi tushunchasi. Zaruriy shart.

2⁰. Funktsiya ekstremumga erishishining yetarli sharti.

1⁰. Funktsiya ekstremumi tushunchasi. Zaruriy shart.

Faraz qilaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funktsiya $E \subset R^m$ to'plamda berilgan bo'lib, $x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in E$ bo'lsin.

1-ta'rif. Agar shunday $\delta > 0$ son topilsaki,

$$U_\delta(x^0) \subset E \text{ bo'lib, } \forall x \in U_\delta(x^0) \text{ da } f(x) \leq f(x^0)$$

bo'lsa, $f(x)$ funktsiya x^0 nuqtada lokal maksimumga, $f(x) \geq f(x^0)$ bo'lsa, $f(x)$ funktsiya x^0 nuqtada lokal minimumga erishadi deyiladi.

2-ta'rif. Agar shunday $\delta > 0$ son topilsaki, $U_\delta(x^0) \subset E$ bo'lib, $\forall x \in U_\delta(x^0) \setminus \{x^0\}$ da $f(x) < f(x^0)$ bo'lsa, $f(x)$ funktsiya x^0 nuqtada qat'iy lokal maksimumga, $f(x) > f(x^0)$ bo'lsa, $f(x)$ funktsiya x^0 nuqtada lokal qat'iy minimumga erishadi deyiladi.

Funktsiyaning lokal maksimumi, lokal minimumi umumiy nom bilan lokal ekstremumi deyiladi. Bunda x^0 nuqta $f(x)$ funktsiyaning lokal ekstremum nuqtasi, $f(x^0)$ ga esa funktsiyaning lokal ekstremum qiymati deyiladi.

Funktsiyaning maksimum (minimum) qiymati quyidagicha belgilanadi:

$$f(x^0) = \max_{x \in U_\delta(x^0)} f(x) \quad \left(f(x^0) = \min_{x \in U_\delta(x^0)} f(x) \right).$$

Ma'lumki,

$$\Delta f(x^0) = f(x) - f(x^0)$$

ayirma $f(x)$ funksiyaning x^0 nuqtadagi to'liq orttirmasi deyilar edi.

$f(x)$ funksiya x^0 nuqtada lokal maksimumga erishsa, unda $\forall x \in U_\delta(x^0)$ da

$$\Delta f(x^0) \leq 0$$

bo'ladi va aksincha.

Shuningdek, $f(x)$ funksiya x^0 nuqtada lokal minimumga erishsa, unda $\forall x \in U_\delta(x^0)$ da

$$\Delta f(x^0) \geq 0$$

bo'ladi va aksincha.

1-teorema. Agar $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtada lokal ekstremumga erishsa va shu nuqtada barcha

$$\frac{\partial f}{\partial x_i} \quad (i = 1, 2, \dots, m)$$

xususiylar hosilalarga ega bo'lsa, u holda

$$\frac{\partial f(x^0)}{\partial x_i} = 0 \quad (i = 1, 2, \dots, m)$$

bo'ladi.

◀ Aytaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtada lokal minimumga erishsin. U holda

$$\forall x = (x_1, x_2, \dots, x_m) \in U_\delta(x^0) \text{ da } f(x_1, x_2, \dots, x_m) \geq f(x_1^0, x_2^0, \dots, x_m^0)$$

tengsizlik bajariladi. Jumladan

$$f(x_1, x_2^0, x_3^0, \dots, x_m^0) \geq f(x_1^0, x_2^0, \dots, x_m^0)$$

bo'ladi. Agar

$$\varphi(x_1) = f(x_1, x_2^0, x_3^0, \dots, x_m^0)$$

deyilsa, $\forall x_1 \in (x_1^0 - \delta, x_1^0 + \delta)$ da

$$\varphi(x_1) \geq \varphi(x_1^0)$$

bo'lib, bir o'zgaruvchili $\varphi(x_1)$ funksiya x_1^0 nuqtada lokal minimumga erishadi. Unda 25-ma'ruzada keltirilgan teorema ko'ra

$$\varphi'(x_1^0)=0, \text{ ya'ni } \frac{\partial f(x^0)}{\partial x_1}=0$$

bo'ladi.

Xuddi shunga o'xshash

$$\frac{\partial f(x^0)}{\partial x_2}=0, \dots, \frac{\partial f(x^0)}{\partial x_m}=0$$

bo'lishi isbotlanadi. ►

1-eslatma. Agar $f(x)$ funksiya biror x^0 nuqtada lokal ekstremumga erishsa va shu nuqtada differensiallanuvchi bo'lsa, u holda

$$df(x^0)=0$$

bo'ladi.

2-eslatma. $f(x)=f(x_1, x_2, \dots, x_m)$ funksiyaning biror x^0 nuqtada barcha xususiy hosilalarga ega va

$$\frac{\partial f(x^0)}{\partial x_i}=0 \quad (i=1, 2, \dots, m)$$

bo'lishidan berilgan funksiyaning shu nuqtada lokal ekstremumga erishishi har doim kelib chiqavermaydi. (misollar keyingi punktda keltiriladi).

Demak, 1-teorema funksiyaning lokal ekstremumga erishishining zaruriy shartini ifodalaydi.

$f(x)$ funksiya xususiy hosilalarini nolga aylantiradigan nuqtalar uning statsionar nuqtalari deyiladi.

2°. Funksiya ekstremumga erishishining yetarli sharti.

Aytaylik, $f(x)=f(x_1, x_2, \dots, x_m)$ funksiya $x^0 \in R^m$ nuqtaning biror $U_\delta(x^0)$ atrofida berilgan, shu atrofda barcha ikkinchi tartibli uzluksiz xususiy hosilalarga ega va

$$\frac{\partial f(x^0)}{\partial x_i} = 0 \quad (i=1,2,\dots,m)$$

bo'lsin. Bu funksiyaning Teylor formulasi (62-ma'ruzada keltirilgan Teylor formulasida $n=2$ bo'lgan hol),

$$\frac{\partial f(x^0)}{\partial x_i} = 0 \quad (i=1,2,\dots,m)$$

shartni hisobga olgan holda, quyidagicha

$$f(x) = f(x^0) + \frac{1}{2} \sum_{i,k=1}^m \frac{\partial^2 f}{\partial x_i \partial x_k} \Delta x_i \Delta x_k \quad (1)$$

bo'ladi, bunda ikkinchi tartibli xususiy hosilalar

$$(x_1^0 + \theta \cdot \Delta x_1, x_2^0 + \theta \cdot \Delta x_2, \dots, x_m^0 + \theta \cdot \Delta x_m)$$

$(0 < \theta < 1)$ nuqtada hisoblangan va

$$\Delta x_1 = x_1 - x_1^0, \Delta x_2 = x_2 - x_2^0, \dots, \Delta x_m = x_m - x_m^0.$$

Berilgan $f(x)$ funksiya ikkinchi tartibli xususiy hosilalarning statsionar nuqta x^0 dagi qiymatlarini

$$a_{ik} = \frac{\partial^2 f(x^0)}{\partial x_i \partial x_k} \quad (i, k=1,2,\dots,m)$$

bilan belgilaymiz. Barcha ikkinchi tartibli xususiy hosilalar

$$\frac{\partial^2 f}{\partial x_i \partial x_k}$$

larning $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtada uzluksizligidan

$$a_{ik} = a_{ki}$$

hamda

$$\frac{\partial^2 f(x_1^0 + \theta \Delta x_1, x_2^0 + \theta \cdot \Delta x_2, \dots, x_m^0 + \theta \cdot \Delta x_m)}{\partial x_i \partial x_k} = \frac{\partial^2 f(x^0)}{\partial x_i \partial x_k} + \alpha_{ik} = a_{ik} + \alpha_{ik}$$

bo'lishi kelib chiqadi, bunda

$$\Delta x_i \rightarrow 0 \quad (i=1,2,\dots,m) \text{ da } \alpha_{ik} \rightarrow 0.$$

Natijada (1) tenglik ushbu

$$\Delta f(x^0) = f(x) - f(x^0) = \frac{1}{2} \left[\sum_{i,k=1}^m a_{ik} \Delta x_i \Delta x_k + \sum_{i,k}^m \alpha_{ik} \Delta x_i \Delta x_k \right]$$

ko‘rinishga keladi.

Agar

$$\rho = \sqrt{\Delta x_1^2 + \Delta x_2^2 + \dots + \Delta x_m^2},$$

$$\Delta x_i = \rho \cdot \xi_i \quad (i=1, 2, \dots, m)$$

deyilsa, so‘ng $\Delta x_i \rightarrow 0$ ($i=1, 2, \dots, m$) da, ya‘ni $\rho \rightarrow 0$ da

$$\sum_{i,k=1}^m \alpha_{ik} \Delta x_k \Delta x_i = \rho^2 \sum_{i,k}^m \alpha_{ik} \xi_i \xi_k = \rho^2 \cdot \alpha(\rho)$$

(bunda, $\rho \rightarrow 0$ da $\alpha(\rho) \rightarrow 0$) bo‘lishini e‘tiborga olsak, u holda

$$\Delta f(x^0) = \frac{\rho^2}{2} \left[\sum_{i,k=1}^m a_{i,k} \xi_i \xi_k + \alpha(\rho) \right] \quad (2)$$

bo‘lishini topamiz.

Ma‘lumki, $\Delta f(x^0) = f(x) - f(x^0)$ ayirma $U_\delta(x^0)$ da ishora saqlasa, ya‘ni $\forall x \in U_\delta(x^0)$ da

$$\Delta f(x^0) \geq 0$$

bo‘lsa, $f(x)$ funksiya x^0 nuqtada lokal minimumga,

$$\Delta f(x^0) \leq 0$$

bo‘lsa, $f(x)$ funksiya x^0 nuqtada lokal maksimumga erishadi.

Yuqoridagi (2) tenglikdan ko‘rinadiki, $\Delta f(x^0)$ ning ishorasi koeffitsientlari

$$a_{ik} = \frac{\partial^2 f(x^0)}{\partial x_i \partial x_k} \quad (i, k = 1, 2, \dots, m)$$

bo‘lgan

$$\sum_{i,k=1}^m a_{ik} \xi_i \xi_k \quad (3)$$

kvadratik formaga bog‘liq bo‘ladi.

2-teorema. Agar (3) kvadratik forma musbat aniqlangan bo'lsa, $f(x)$ funksiya x^0 nuqtada lokal minimumga, manfiy aniqlangan bo'lsa, lokal maksimumga erishadi.

Agar (3) kvadratik forma noaniq bo'lsa, $f(x)$ funksiya x^0 nuqtada lokal ekstremumga erishmaydi.

◀ Bu teorema, keyingi punktda, xususiy holda ya'ni ikki o'zgaruvchili funksiyalar uchun isbotlanadi. ▶ (qaralsin, [1], 13-bob)

Xususiy hollar. $m=1$ bo'lsin. bu holda $u = f(x) \quad (x \in R, u \in R)$ funksiyaning lokal ekstremumlari, ekstremumning zaruriy va yetarli shartlari kabi tushuncha va tasdiqlarga kelimiz. Ular 25-ma'ruzada bayon etilgan.

$m=2$ bo'lsin. Bu holda $u = f(x, y) \quad ((x, y) \in R^2, u \in R)$ ikki o'zgaruvchili funksiyaning lokal ekstremum tushunchalari yuzaga kelib, bu hol uchun ularning ta'riflari quyidagicha bo'ladi.

Aytaylik, $u = f(x, y)$ funksiya $E \subset R^2$ to'plamda berilgan bo'lib, $(x_0, y_0) \in E$ bo'lsin.

Agar shunday $\delta > 0$ son topilsaki, $U_\delta((x_0, y_0)) \subset E$ bo'lib, $\forall (x, y) \in U_\delta((x_0, y_0))$ uchun

$$f(x, y) \geq f(x_0, y_0) \quad (f(x, y) \leq f(x_0, y_0))$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada lokal minimumga (lokal maksimumga) erishadi deyiladi. (x_0, y_0) nuqta $f(x, y)$ funksiyaning lokal minimum (maksimum) nuqtasi, $f(x_0, y_0)$ miqdor esa funksiyaning minimum (maksimum) qiymati deyiladi.

Agar shunday $\delta > 0$ son topilsaki, $U_\delta((x_0, y_0)) \subset E$ bo'lib, $\forall (x, y) \in U_\delta((x_0, y_0)) \setminus \{(x_0, y_0)\}$ uchun

$$f(x, y) > f(x_0, y_0) \quad (f(x, y) < f(x_0, y_0))$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada qat'iy lokal minimumga (qat'iy lokal maksimumga) erishadi deyiladi.

1-misol. Ushbu

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

funksiyaning $(0,0)$ nuqtada qat'iy maksimumga erishishi ko'rsatilsin.

◀ $\delta > 0$ ($0 < \delta < 1$) sonni olib, $(0,0)$ nuqtaning $U_\delta((0,0))$ atrofini hosil qilamiz.

Unda $\forall (x, y) \in U_\delta((0,0)) \setminus \{(0,0)\}$ uchun

$$f(x, y) = \sqrt{1 - x^2 - y^2} < f(0,0) = 1$$

bo'ladi. Demak, berilgan funksiya $(0,0)$ nuqtada maksimumga erishadi. ▶

Agar $f(x, y)$ funksiya $(x_0, y_0) \in E \subset \mathbb{R}^2$ nuqtada lokal ekstremumga erishsa va shu nuqtada

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$$

xususiy hosilalarga ega bo'lsa, u holda

$$\frac{\partial f(x_0, y_0)}{\partial x} = 0, \quad \frac{\partial f(x_0, y_0)}{\partial y} = 0$$

bo'ladi.

Biroq, $f(x, y)$ funksiyaning biror (x^*, y^*) nuqtada $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ xususiy hosilalari mavjud bo'lib, ular shu nuqtada nolga teng bo'lsa, qarayotgan funksiya (x^*, y^*) nuqtada ekstremumga erishmasdan qolishi mumkin. Masalan,

$$f(x, y) = xy$$

funksiya

$$\frac{\partial f}{\partial x} = y, \quad \frac{\partial f}{\partial y} = x$$

xususiy hosilalarga ega bo'lib, ular $(0,0)$ nuqtada nolga teng:

$$\frac{\partial f(0,0)}{\partial x} = 0, \quad \frac{\partial f(0,0)}{\partial y} = 0$$

bo'lsa ham, bu funksiya $(0,0)$ nuqtada ekstremumga erishmaydi (funksiya grafigi-giperbolik paraboloidni tasavvur qiling).

Aytaylik, $f(x, y)$ funksiya $(x_0, y_0) \in R^2$ nuqtaning biror $U_\delta((x_0, y_0))$ atrofida ($\delta > 0$) berilgan bo'lib, quyidagi shartlarni bajarsin:

- 1) $f(x, y)$ funksiya $U_\delta((x_0, y_0))$ da uzulksiz va uzluksiz $f'_x, f'_y, f''_{x^2}, f''_{xy}, f''_{y^2}$ xususiy hosilalarga ega,
- 2) (x_0, y_0) statsionar nuqta:

$$f'_x(x_0, y_0) = 0, \quad f'_y(x_0, y_0) = 0.$$

Bu $f(x)$ funksiya uchun 2^0 da yuritilgan mulohazalarni qo'llab

$$\begin{aligned} \Delta f(x_0, y_0) &= f(x, y) - f(x_0, y_0) = \\ &= \frac{1}{2} (a_{11} \Delta x^2 + 2a_{12} \Delta x \Delta y + a_{22} \Delta y^2 + \alpha_{11} \Delta x^2 + 2\alpha_{12} \Delta x \Delta y + \alpha_{22} \Delta y^2) \end{aligned} \quad (*)$$

bo'lishini topamiz, bunda

$$a_{11} = f''_{x^2}(x_0, y_0), \quad a_{12} = f''_{xy}(x_0, y_0) = f''_{yx}(x_0, y_0), \quad a_{22} = f''_{y^2}(x_0, y_0)$$

bo'lib,

$$\Delta x \rightarrow 0, \quad \Delta y \rightarrow 0 \quad \partial a \quad \alpha_{11} \rightarrow 0, \quad \alpha_{12} \rightarrow 0, \quad \alpha_{22} \rightarrow 0$$

bo'ladi.

3-teorema. Agar

$$a_{11} \Delta x^2 + 2a_{12} \Delta x \Delta y + a_{22} \Delta y^2 \quad (4)$$

kvadratik forma musbat aniqlangan, ya'ni

$$a_{11} > 0, \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} a_{22} - a_{12}^2 > 0$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada lokal minimumga erishadi, agar (4)

kvadratik forma manfiy aniqlangan, ya'ni

$$a_{11} < 0, \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} a_{22} - a_{12}^2 > 0$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada lokal maksimumga erishadi.

◀ Ma'lumki, $f(x, y)$ funksiyaning (x_0, y_0) nuqtada ekstremumga erishishi $U_\delta((x_0, y_0))$ da ushbu

$$\Delta f(x_0, y_0) = f(x, y) - f(x_0, y_0)$$

ayirmaning ishora saqlashi bilan bog'liq:

$$\forall (x, y) \in U_\delta((x_0, y_0)) \text{ da } \Delta f(x_0, y_0) > 0 \text{ bo'lsa,}$$

(x_0, y_0) nuqtada lokal minimum, $\Delta f(x_0, y_0) < 0$ bo'lsa. (x_0, y_0) nuqtada lokal maksimum sodir bo'ladi.

$\Delta f(x_0, y_0)$ ayirmaning ishorasini aniqlash qulay bo'lishi maqsadida (4) da

$$\Delta x = \rho \cdot \cos \varphi, \Delta y = \rho \cdot \sin \varphi$$

almashtirishni bajaramiz, bunda

$$\rho = \sqrt{\Delta x^2 + \Delta y^2}.$$

Natijada (*) munosabat ushbu

$$\begin{aligned} \Delta f(x_0, y_0) = & \frac{\rho^2}{2} \left[(a_{11} \cos^2 \varphi + 2a_{12} \cos \varphi \sin \varphi + a_{22} \sin^2 \varphi) + \right. \\ & \left. + (a_{11} \cos^2 \varphi + 2a_{12} \cos \varphi \sin \varphi + a_{22} \sin^2 \varphi) \right] \end{aligned} \quad (5)$$

ko'rinishga keladi.

Aytaylik,

$$a_{11} > 0, \quad a_{11}a_{22} - a_{12}^2 > 0$$

bo'lsin.

Ravshanki,

$$\begin{aligned} & a_{11} \cos^2 \varphi + 2a_{12} \cos \varphi \sin \varphi + a_{22} \sin^2 \varphi = \\ & = \frac{1}{a_{11}} \left[(a_{11} \cos \varphi + a_{12} \sin \varphi)^2 + (a_{11}a_{22} - a_{12}^2) \cdot \sin^2 \varphi \right]. \end{aligned}$$

Ayni paytda, bu funksiya, φ ning funksiyasi sifatida $[0, 2\pi]$ da uzluksiz bo'lib, o'zining eng kichik qiymati (uni m bilan belgilaylik) m ga ega bo'ladi:

$$\left| a_{11} \cos^2 \varphi + 2a_{12} \cos \varphi \sin \varphi + a_{22} \sin^2 \varphi \right| \geq m > 0.$$

Ikkinchi tomondan, $\Delta x \rightarrow 0$, $\Delta y \rightarrow 0$ ya'ni $\rho \rightarrow 0$ da $\alpha_{11} \rightarrow 0$, $\alpha_{12} \rightarrow 0$, $\alpha_{22} \rightarrow 0$ bo'lganligi sababli, ρ ning etardli kichik qiymatlarida

$$|\alpha_{11} \cos^2 \varphi + 2\alpha_{12} \cos \varphi \sin \varphi + \alpha_{22} \sin^2 \varphi| \leq |\alpha_{11}| + 2|\alpha_{12}| + |\alpha_{22}| < m$$

bo'laoladi.

Demak, $a_{11} > 0$, $a_{11} a_{22} - a_{12}^2 > 0$ bo'lganda (5) tenglikning o'ng tomonidagi ifoda musbat bo'ladi. Binobarin,

$$\Delta f(x_0, y_0) > 0$$

bo'lib, $f(x, y)$ funksiya (x_0, y_0) nuqtada lokal minimumga erishadi.

Aytaylik,

$$a_{11} < 0$$
 , $a_{11} a_{22} - a_{12}^2 > 0$

bo'lsin. Bu holda (5) tenglikning o'ng tomonidagi ifoda manfiy bo'ladi. Binobarin,

$$\Delta f(x_0, y_0) < 0$$

bo'lib, $f(x, y)$ funksiya (x_0, y_0) nuqtada lokal maksimumga erishadi. ►

3-eslatma. Agar

$$a_{11} a_{22} - a_{12}^2 < 0$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada ekstremumga erishmaydi.

4-eslatma. Agar

$$a_{11} a_{22} - a_{12}^2 = 0$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada ekstremumga erishishi ham mumkin, erishmasligi ham mumkin (qaralsin, [1], 13-bob).

2-misol. Ushbu

$$f(x, y) = x^2 + xy + y^2 - 2x - 3y$$

funksiya ekstremumga tekshirilsin.

◀ Avvalo berilgan funksiyaning statsionar nuqtalarini topamiz:

$$f'_x(x,y)=2x+y-2, \quad 2x+y-2=0, \quad x_0=\frac{1}{3}$$

$$f'_y(x,y)=x+2y-3, \quad x+2y-3=0, \quad y_0=\frac{4}{3}.$$

Demak, $\left(\frac{1}{3}, \frac{4}{3}\right)$ statsionar nuqta.

Ravshanki,

$$f''_{x^2}(x,y)=2, \quad f''_{xy}(x,y)=1, \quad f''_{y^2}(x,y)=2.$$

Demak,

$$a_{11}=2, \quad a_{12}=1, \quad a_{22}=2$$

$a_{11}=2>0$ va $a_{11}a_{22}-a_{12}^2=3>0$ bo'lganligi uchun berilgan funksiya $\left(\frac{1}{3}, \frac{4}{3}\right)$ nuqtada lokal minimumga erishadi va

$$\min f(x,y)=f\left(\frac{1}{3}, \frac{4}{3}\right)=-\frac{7}{3}$$

bo'ladi. ►

3-misol. Ushbu

$$f_1(x,y)=x^4+y^4,$$

$$f_2(x,y)=-(x^4+y^4),$$

$$f_3(x,y)=x^3+y^3,$$

funksiyalar ekstremumga tekshirilsin.

◄ Berilgan funksiyalar uchun $(0,0)$ statsionar nuqta bo'ladi. Bu funksiyalar uchun

$$a_{11}a_{22}-a_{12}^2=0$$

bo'ladi. Ravshanki, $(0,0)$ nuqtada $f_1(x,y)$ funksiya minimumga, $f_2(x,y)$ funksiya esa maksimumga erishadi. $f_3(x,y)$ funksiya $(0,0)$ nuqtada ekstremumga ega bo'lmaydi. ►

Adabiyotlar

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